THE PHYSICAL AI CO-DESIGN MATRIX

Twenty-five questions. Five cognitive obligations against five physical constraints. Part II answers them one row at a time.

	C1 Time & Freshness	C2 Inertia & Stopping	C3 Actuation Limits	C4 Energy & Thermal	C5 Silicon Determinism
PERCEIVE Ch 4 · OBS-01	When did this observation happen?	How far ahead must you see?	*not binding* commands nothing	What does sensor heat do to geometry?	**DOMINANT** What does moving the pixels cost?
REMEMBER Ch 5 · STATE-01	**DOMINANT** What is a belief worth after Δt?	How much world must you hold?	*thin* relocalization snap	Which calibration drifts with heat?	Bounded state without allocation
REASON Ch 6 · INTENT-01	**DOMINANT** How long may intent outlive its world?	*not binding* reaches via the lease	*not binding* commands no torque	What does throttling do to a deadline?	Keeping a large model off the real-time core
PLAN Ch 7 · PLAN-01	Outliving the latency that produced it	What makes a plan abandonable?	**DOMINANT** What shapes may a command take?	What does a motion cost per shift?	*thin* compute as a budget
EXECUTE Ch 8 · ENF-01	The deadline, and the cycle it is missed	**DOMINANT** How much margin does stopping need?	Can the veto violate the limit it protects?	What clamps the current asked for?	Code that may never pause

READING THE GRID: **Bold border** marks each chapter's dominant cell, where most of its length goes. **Shaded** cells do not bind, and saying why is part of the argument. The grid is uneven because the physics is.